

The University of Texas at Dallas
CS 6320 Natural Language Processing
Spring 2025

Homework 1: 100 points
Issued Feb 3rd, 2025
Due Feb 17th, 2025 midnight

Student: Khoi Nguyen (kxn2200220)

Problem 1: Regular Expressions

1.1 Write regular expressions for the following languages. (2 points each)

1. Strings starting with one or more lowercase letters (a-z), followed by a space character and then two to four digits.

`^[a-z]+ \d{2-4}`

2. Strings starting with an uppercase letter, followed by at least one lowercase letter, like “Apple”, “Banana”, or “Carrot”.

`^[A-Z][a-z]+$`

3. Strings (sequence of letters, digits, or underscores) with two consecutive repeated words (e.g., “Humbert Humbert” and “th_e th_e” but not “the bug” or “the big bug”);

`([\w\d\_]+){2}`

4. Strings (sequence of letters, digits, or underscores) that have both the word “hedge” and the word “fund” in them (but not, e.g., words like funds that merely contain the word fund);

Solution 1:

`([\w\d\_]+)(\bhedge\b)([\w\d\_]+)(\bfund\b)([\w\d\_]+) | ([\w\d\_]+)(\bfund\b)([\w\d\_]+)(\bhedge\b)([\w\d\_]+)`

])+)

Solution 2: $^((?=.*\backslash\text{b}\text{hedge}\backslash\text{b})(?=.*\backslash\text{b}\text{fund}\backslash\text{b})[\backslash\text{w}])+\$$

5. All binary strings except the empty string (sequence of 0s or 1s).

$^([01])+\$$

6. Binary strings containing at least three 1s.

$([01]^*1[01]^*)\{3\}$

7. Binary strings of odd length.

$^([01]\{2\})^*[01]\$$

8. Binary strings that start and end with the same character.

$(0[01]^*0)|(1[01]^*1)$

9. Binary strings with the number of 0s as a multiple of 3.

$^((1^*01^*)\{3\})^*\$$

10. Binary strings start with 0 and have an odd length, or start with 1 and have an even length.

$^0([01]\{2\})^*|1([01]\{2\})^*[01]\$$

11. All strings of as and bs which contain the substring abba.

$^[ab]^*abba[ab]^*\$$

12. All strings of xs and ys where every y is immediately followed by at least three xs.

$^(x^*yx\{3\}x^*)^*\$$

13. Which of the following strings is not matched by the regular expression

$(b(ab|a)^*d)^*$

- A. bd
- B. babd
- C. baadbaabdbad

D. Baadbaabbdbad ← this one

14. Which of the following matches the regular expression

$(,\backslash s[0-9]\{1,4\})\{4\},\backslash s[0-9]\{1,3\}\backslash .[0-9]$

A. , 135, 1155, 915, 513, 18.8 ← this one

B. , 135, 11557, 915, 513, 18.8

C. , 135, 1155, 915, 513, 188

15. How many strings of length less than 4 contain the language described by the regular expression $(x|y)^*y(a|ab)^*$?

A. 7

B. 10

C. 12 ← this many

D. 11

1.2 Programming exercise. (10 points)

See attached hw1_p1.py.

Problem 2: Simple Text Classification (20 points)

2.1 Recall the simple Naive Bayes text classifier. Why do we need the prior of class distribution? What will happen if we remove this prior? You may use some examples to explain. (10 points)

We need the prior of class distribution because that will help incorporating existing knowledge about the classes to the classifier. That will be most helpful when there is little training data or the dataset is unbalanced.

For example, if examples in class C always have the word W. That does not mean any text with the word W belongs to that class C. We need to multiply $P(W|C)$ with $P(C)$.

2.2 Give two limitations of the simple Naive Bayes text classifier, you may use some examples to explain. (10 points)

Some limitations are:

- No notion of word ordering: “the movie is so good, not bad at all” vs “the movie is bad, not good at all” has the same BOW representation, thus the same NB prediction. Note that this depends on the feature.
- It assigns very low probability to longer sequences. For example, $P(\text{“United States”})$ will be smaller than $P(\text{“United States of America”})$, but they should be equally probable.

Problem 3: N-gram Language Models (40 points)

*Code: see attached **hw1_p3.py***

1. Sentence probability without smoothing:

0.0

2. Sentence probability with smoothing:

1.7204128298897112e-25

3. Predicted next 5 words:

million dollars . </s>

(it ends early)

Appendix

In problem 3:

- Details about the Brown corpus:
 - 57340 sentences
 - num tokens: 20.25 average, 18 median, 180 max, 1 min; about 1M tokens totally
 - vocab size: 49817